

Tim Givney

Industrial Engineering • Manufacturing • Technical Systems

Sydney, NSW Australia | timgivney@gmail.com | 0432504302

Professional Summary

I design, manufacture, and field-test high-reliability mechanical and electronic systems that survive the harshest real-world environments. From heavy-industry mine sites to precision CNC machining and technical media with millions of views, I combine hardcore workshop execution with advanced engineering to turn complex concepts into working hardware.

Key Metrics:

- 2M+ YouTube Views on technical content
 - 15+ Years of Engineering Experience
 - Expertise across industrial manufacturing, CNC machining, electronics, and embedded systems
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Experience

PartsBender

Engineering & Operations Support | 2023 – Present

Supporting industrial engineering and manufacturing operations across product development, CAD systems, production preparation, inventory coordination, and technical e-commerce.

Cobalt CNC

CNC Workshop Operator | 2023 – 2024

Operated SYIL X11 CNC systems for precision machining and fabrication. Utilised 3D scanning technologies and CAD workflow support for product development.

Core Electronics

Creative Technologist / Engineer | 2020 – 2023

Produced 100+ educational videos and technical resources for the maker community, accumulating 2M+ YouTube views. Facilitated workshops and technical communication.

CASMAT Pty Ltd

Contract Engineer | 2019 – 2020

Executed critical industrial maintenance projects within high-stakes, heavy-industry sites including the Tomago Aluminium Smelter and Liddell Power Station under strict safety compliance.

Coal Mines (Mount Thorley & Hunter Valley)

Industrial Machinery Servicing | 2018

Handled heavy machinery asset maintenance at Mount Thorley and Hunter Valley Coal Mines, ensuring operational reliability in demanding industrial environments.

Featured Projects

AP+ AP50 Diaphragm Pump

Lead Designer & Product Development

Precision vacuum diaphragm pump designed for industrial applications. Engineered high-performance 50CFM pump with optimized casting, FEA analysis, and precision manufacturing workflows.

AP+ 2C440 Air Compressor

Design & Development

Advanced air compressor system with precision engineering and thermal optimization. Developed through iterative testing and FEA analysis to deliver high-performance industrial compression.

SKAPA Sunglasses

Product Design & Manufacturing

High-performance sunglasses engineered for durability and style. Designed with precision manufacturing techniques and optimized for comfort and visual clarity in demanding environments.

Long Range Radio (LoRa) Weather and Air Quality Station

IoT & Environmental Monitoring

A battery-powered long-range weather and air quality monitoring station built with a Raspberry Pi Pico and LoRaWAN, capable of transmitting environmental data to a live online dashboard for remote monitoring and alerts.

WLED Smart Lighting

Technical Education & IoT

How to Easily Control Addressable LEDs with an ESP32 or ESP8266 | WLED Project. Comprehensive guide reaching 600k+ views, demonstrating custom pixel art and LED effects without coding.

Technical Capabilities & Expertise

Mechanical Engineering & Industrial Manufacturing

- Design & Reverse Engineering

- Machining & Fabrication
- Fluid Flow & Simulation
- Structural Analysis

Hardware, Firmware & Intelligent Systems

- Microcontrollers & Hardware
- Firmware & Automation
- IoT & Integration

Software, CAD & Engineering Tools

- Design & Simulation
- Programming & Systems
- Operations & E-Commerce

Technical Communication & Operations

- Engineering Media
- Documentation
- Leadership & Compliance

Commercial Media & Industrial Imaging

- Visual Asset Creation
- Corporate & Site Media
- Post-Production & Editing

Leadership & Volunteer Work

Save Our Coast

Board Director & Treasurer

Driving environmental conservation and coastal protection initiatives. Responsible for strategic direction, financial management, and community engagement to protect Australia's marine ecosystems.

Education & Certifications

University of Sydney

Bachelor of Engineering (Mechanical Engineering)

Professional Certifications

- Forklift Licence
 - Working at Heights & Confined Spaces
 - White Card Construction Certification
 - Lifesaver Certification (Merewether Beach)
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Contact

Email: timgivney@gmail.com

Phone: 0432504302

Location: Sydney, NSW Australia

Whether it's manufacturing, product development, technical systems, or solving real-world engineering problems, I'm always interested in meaningful projects and new challenges.